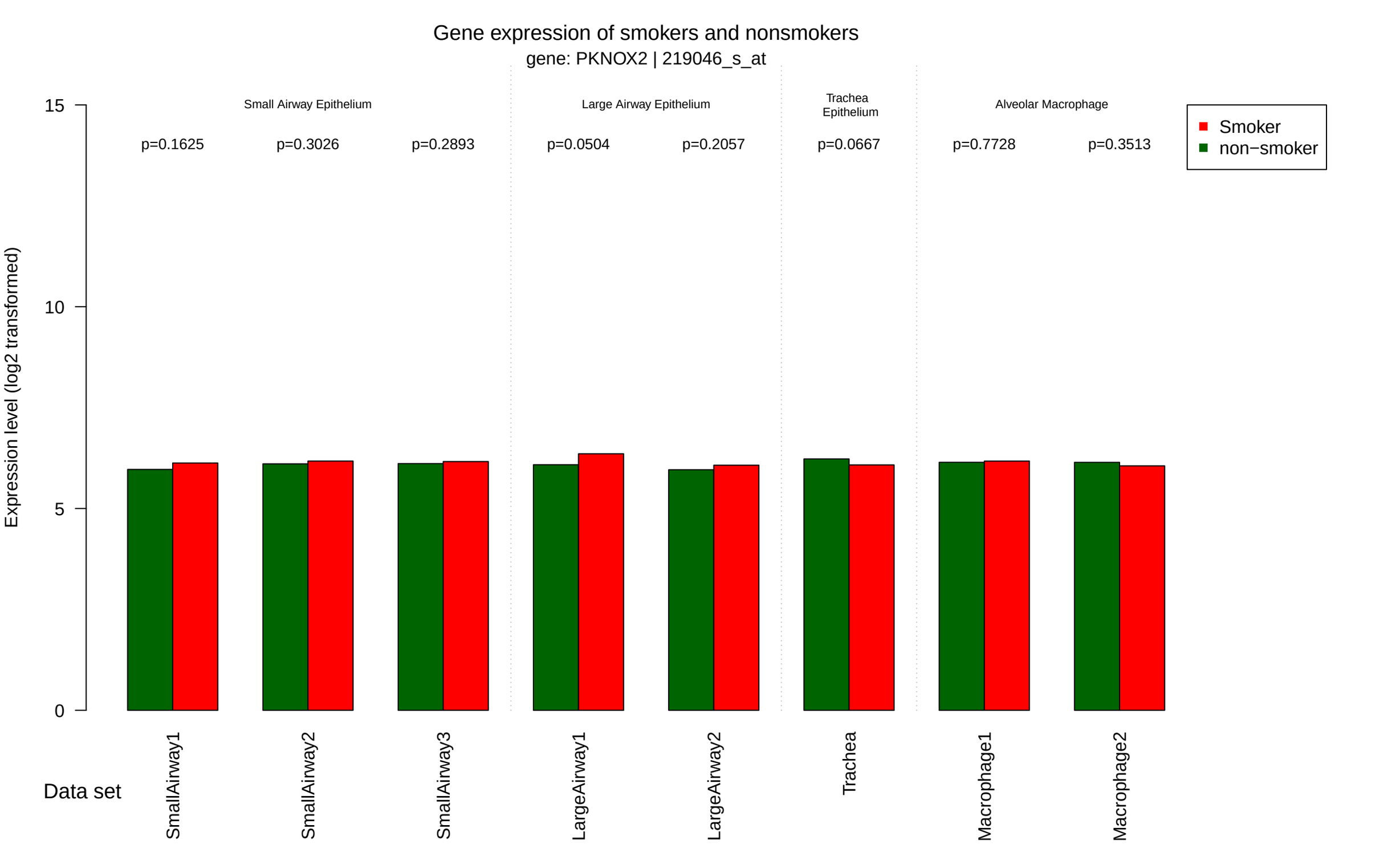
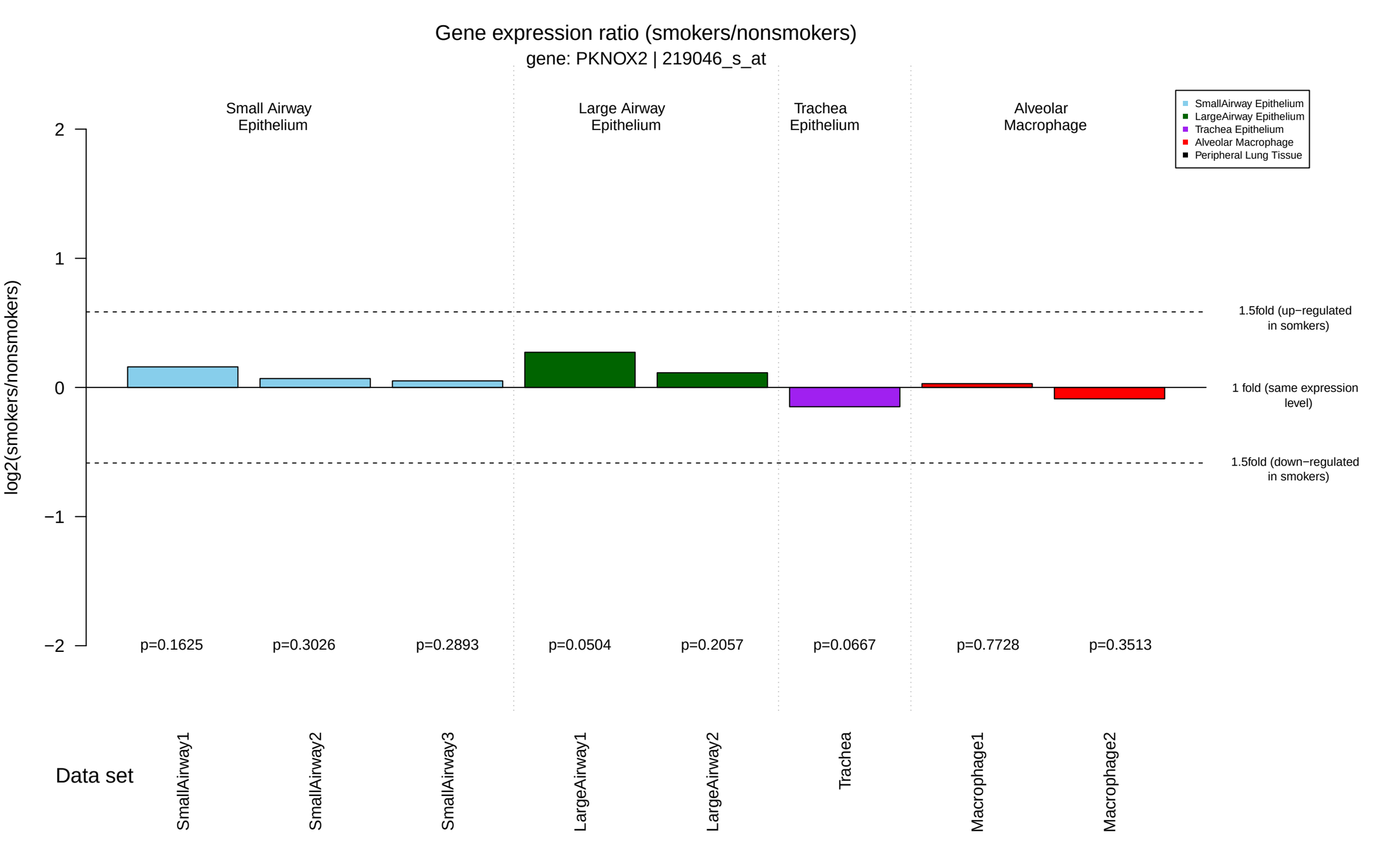
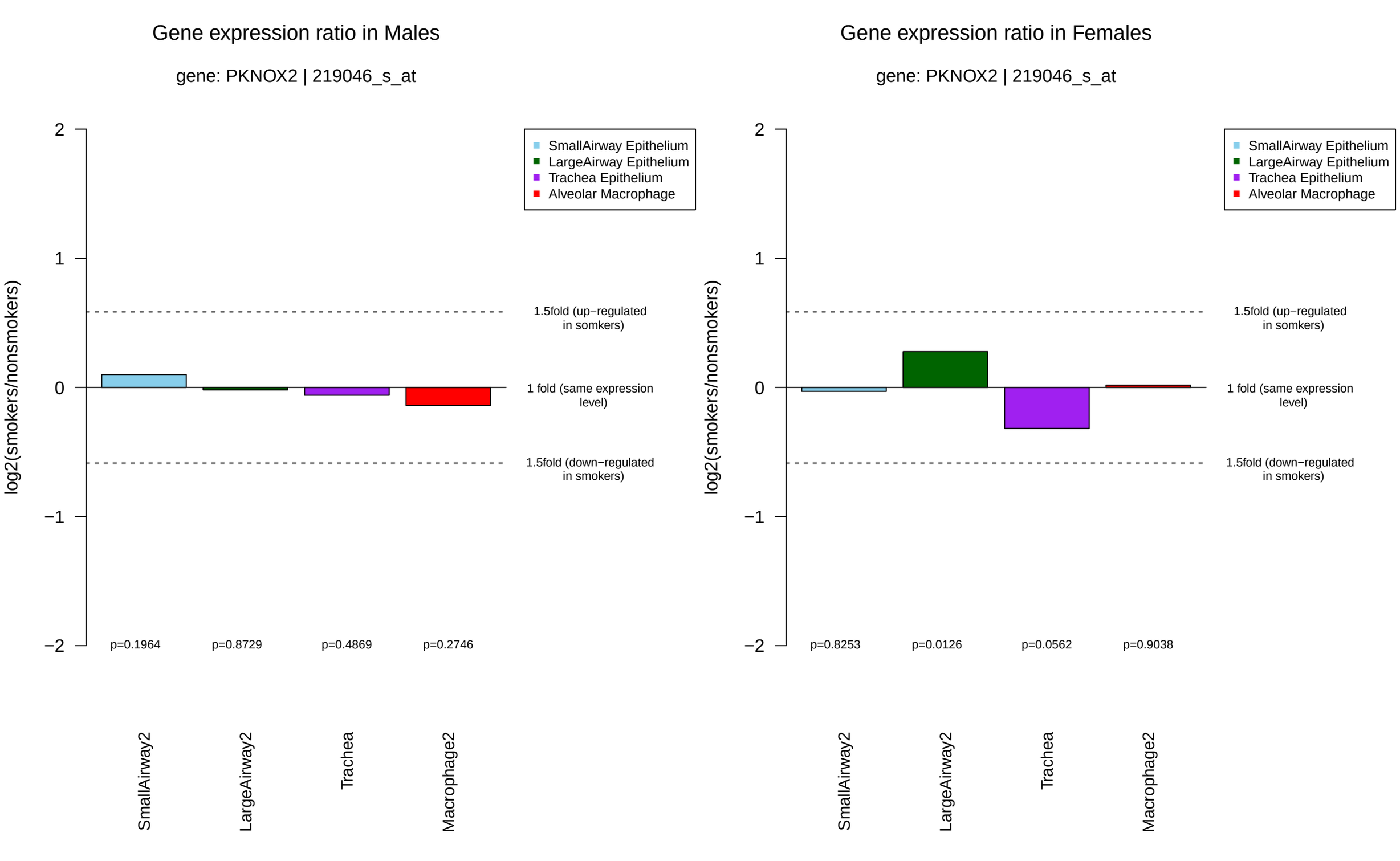
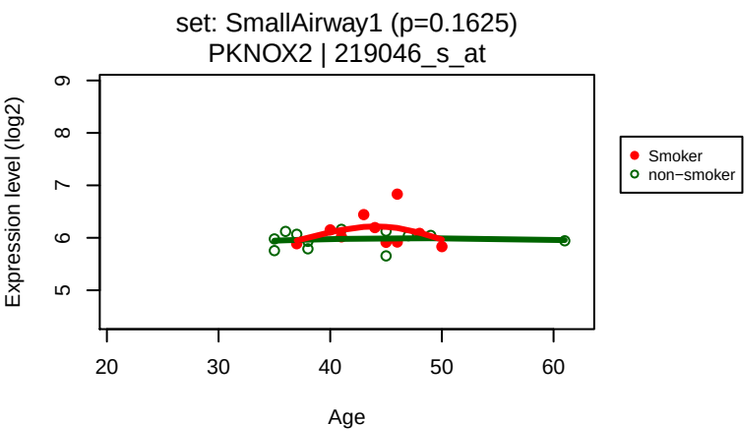
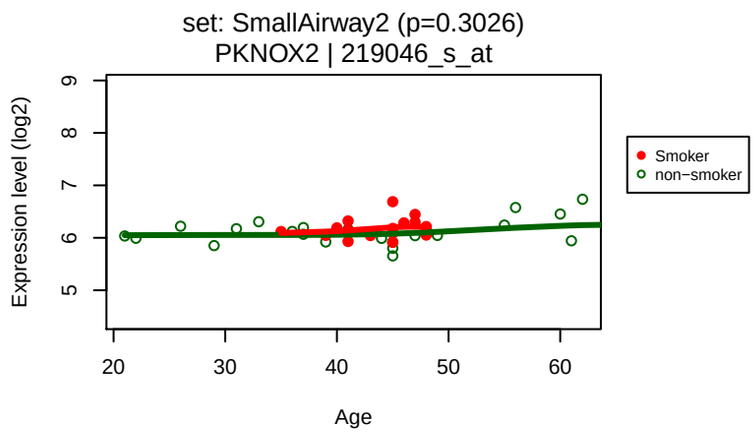
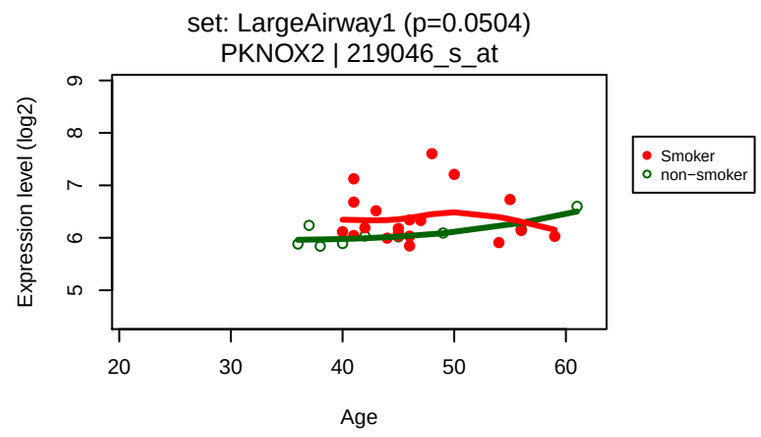

Gene expression of smokers and nonsmokers
gene: PKNOX2 | 219046_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1625
p=0.3026
p=0.2893
p=0.0504
p=0.2057
p=0.0667
p=0.7728
p=0.3513
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

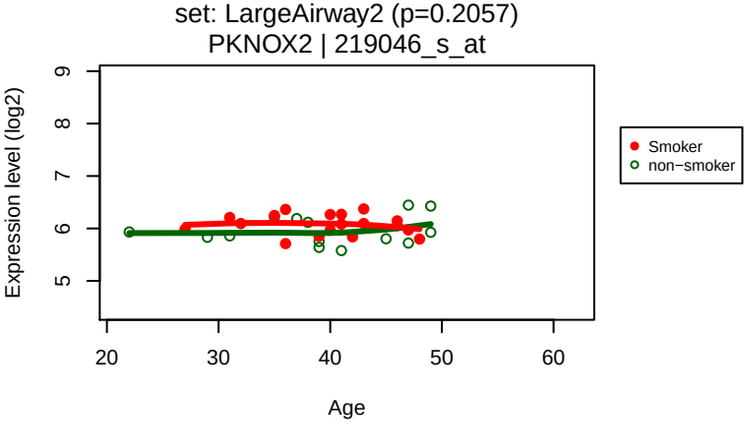

Gene expression ratio (smokers/nonsmokers)
gene: PKNOX2 | 219046_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1625
p=0.3026
p=0.2893
p=0.0504
p=0.2057
p=0.0667
p=0.7728
p=0.3513
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

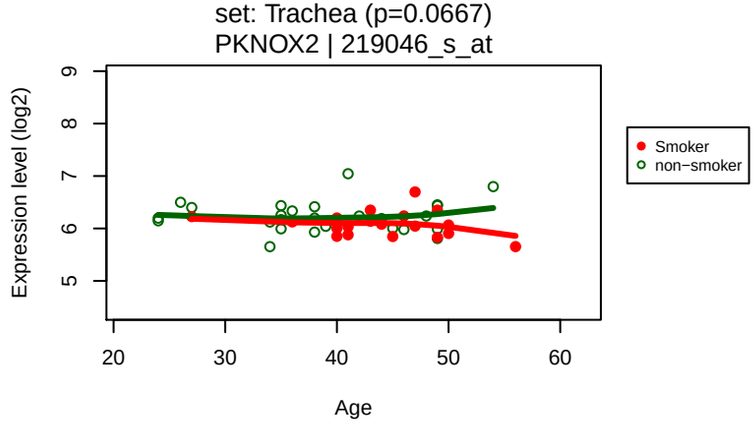

Gene expression ratio in Males
gene: PKNOX2 | 219046_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1964
p=0.8729
p=0.4869
p=0.2746
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PKNOX2 | 219046_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8253
p=0.0126
p=0.0562
p=0.9038
SmallAirway2
LargeAirway2
Trachea
Macrophage2

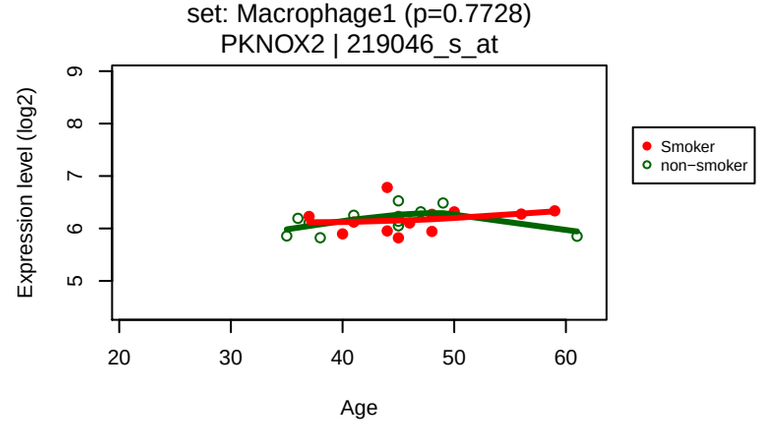

set: SmallAirway1 (p=0.1625)
PKNOX2 | 219046_s_at
Expression level (log2)
Age
Smoker
non−smoker

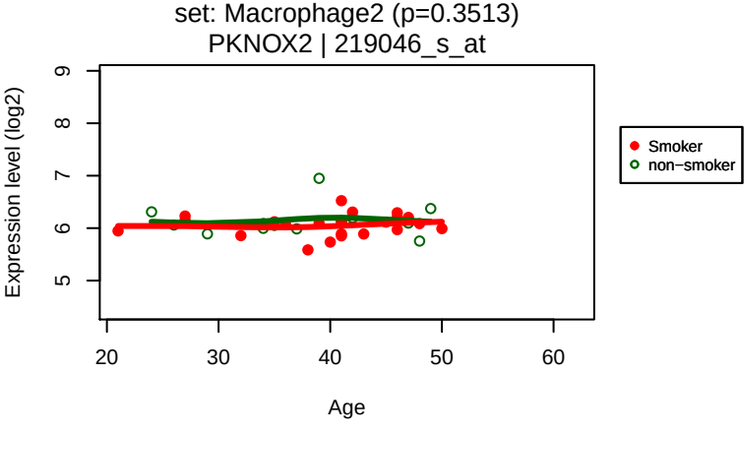

set: SmallAirway2 (p=0.3026)
PKNOX2 | 219046_s_at
Expression level (log2)
Age
Smoker
non−smoker

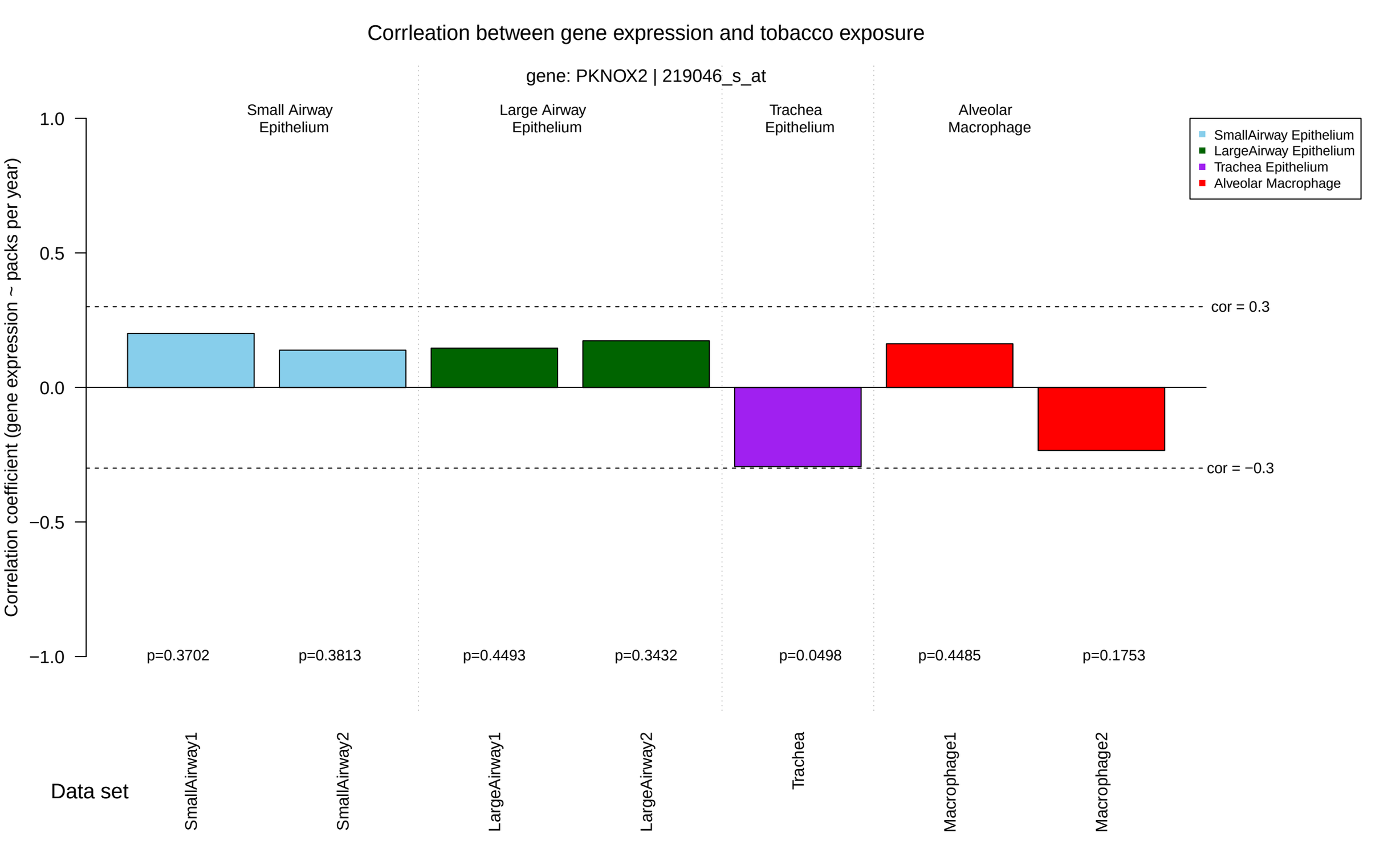

set: LargeAirway1 (p=0.0504)
PKNOX2 | 219046_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2057)
PKNOX2 | 219046_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0667)
PKNOX2 | 219046_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7728)
PKNOX2 | 219046_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3513)
PKNOX2 | 219046_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PKNOX2 | 219046_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3702
p=0.3813
p=0.4493
p=0.3432
p=0.0498
p=0.4485
p=0.1753
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2